Chemosphere 253 (2020) 126687

Contents lists available at ScienceDirect

Chemosphere

journal homepage: www.elsevier.com/locate/chemosphere

Simultaneous H_2S mitigation and methanization enhancement of chicken manure through the introduction of the micro-aeration approach

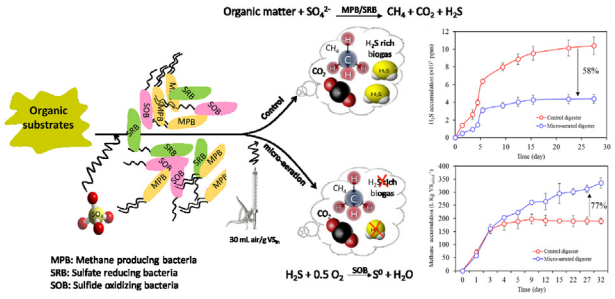
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HIGHLIGHTS

- Micro-aerobic approach was applied to remove/mitigate H_2S of chicken manure biogas.
- Accelerated hydrolysis and lower VFAs content were achieved due to micro-aeration.
- The most efficient micro-aeration trial with 58% removed H_2S was 30 ml air/gVS_{in}.
- Microbial community dynamics were characterized by 16S rDNA cloning library.
- Bacterial richness and diversity were enhanced with micro-aerated digesters.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 20 August 2019
Received in revised form
30 March 2020
Accepted 31 March 2020
Available online 7 April 2020

Handling Editor: Christian Kennes

Keywords:
Micro-aeration
Hydrogen sulfide
Methanogens
Methane yield
Sulfidogenesis

ABSTRACT

The impact on H_2S alleviation and methane yield enhancement after submitting the anaerobic digestion of chicken manure to a finite amount of air was investigated. The largest reduction in the H_2S biogas content (58% lower) occurred when air intensity of 30 ml/g VS_{in} was injected into the reactors. Consequently, a maximum methane yield (335 mL-g VS_{in}^{-1}), which was 77% higher than the control, was concurrently achieved. Slight sulfate accumulation (<330 mg L^{-1}) was observed inside the micro-aerated digesters with higher air intensities, suggesting a suppression of sulfide inhibition. Bacterial diversity/richness was enhanced in these digesters while the relative abundance of *Methanocelleus* increased by 36%. The most important contributing factor to enhancement was the synergistic effect resulting from increments in the hydrolysis rate and the suppression of sulfide inhibition. The results highlighted the potential of *in situ* H_2S mitigation with the added benefit of methane yield enhancement.

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1. Introduction

In the last decade the amount of livestock manure in China has

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https://doi.org/10.1016/j.chemosphere.2020.126687
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rapidly increased, and it is estimated to reach 2.78 billon tons by the year 2020 (Yan et al., 2017). This type of waste is both energy-rich and bio-digestible, making anaerobic digestion (AD) an optimal solution for its management (Anjum et al., 2017; Zhang et al., 2019). However, the high content of sulfurous matter has been identified as a barrier to livestock AD as it can induce the inhibition of anaerobic microbes by H_2S. Methionine, for instance, a sulfur-containing amino acid, is nutritionally essential in feeding regime for both laying hens and broiler chickens to support feather growth and protein synthesis. Thus, chicken manure becomes rich in sulfur-containing amino acids which their anaerobic digestion forms sulfurous compounds. The presence of sulfurous compounds in the AD concert highlights current challenges such as the out-competition among indigenous organisms accompanied by these compounds with methane producers resulting in the decreasing of not only the quality but also the quantity of the biogas. At the same time, the removal of H_2S from biogas makes the whole AD process less economical.

Specifically, sulfate-reducing bacteria (SRB) will be promoted in the presence of organic matters and sulfates under anaerobic condition where sulfate will be used as the terminal electron acceptor and methanogenic substrates (acetate, H_2) will act as electron donors leading to H_2S formation as the main product (Reaction 1 and 2). H_2S is toxic to methanogens at relatively low concentrations. It is also highly reactive, has a bad odour, and is corrosive to piping and equipments associated with the biogas industry. Extra investment costs are thus required in order to maintain the system (Krayzelova et al., 2015). On the other hand, in an AD process, SRB will compete with methanogens to utilize electron and this negatively affects the methane production. Various approaches have been put forward to mitigate the formation of H_2S during the AD process including chemical methods such as precipitation (Dhar et al., 2011), scrubbing (Couvert et al., 2008), oxidation (Khanal and Huang, 2006), pH elevation (Yan et al., 2018), and biological methods (Show et al., 2013). Nevertheless, even if these techniques can remove H_2S from biogas and thus increase the quality and quantity of biogas production, the final economic balance is still not positive.

$$4H_2 + SO_4^{2-} + H^+ \rightarrow HS^- + 4H_2O\ \Delta G^0 = -38.0\ KJ \quad (1)$$

$$CH_3COO^- + SO_4^{2-} \rightarrow HS^- + 2HCO_3^-\ \Delta G^0 = -39.5\ KJ \quad (2)$$

Recent studies have demonstrated that micro-aeration could provide an innovative solution for addressing the H_2S issues (Nguyen and Khanal, 2018). Micro-aeration refers to the addition of a finite amount of air or oxygen into the anaerobic system to support a partial oxidation of H_2S to elemental sulfur by sulfide-oxidizing bacteria (Reaction 3) (Guerrero et al., 2016).

$$H_2S + 0.5\ O_2 \rightarrow S^0 + H_2O\ \Delta G^0 = -209.4\ KJ \quad (3)$$

Micro-aeration could also have a positive impact on the overall AD performance of the process, including hydrolysis efficiency improvement, methane yield enhancement, and the scavenging of hydrogen sulfide; however, other studies were more contradictory. For instance, Botheju et al. (2010) demonstrated an enhancement in the methane yield of starch by 30–55%, and Tertakovsky et al. (2011) increased the methane yield of synthetic wastewater by 26% after adding limited oxygen to anaerobic digesters. However, Ramos et al. (2014) and Diaz et al. (2011) reported a negligible impact of micro-aeration on the process performance of sewage sludge and the methane yield of cellulose, respectively. Overall, a wide range of H_2S removal efficiency, from 30% to more than 99%, has been reported in the literatures (Zhou et al., 2007; Krayzelova et al., 2014; Fdz-Polanco et al., 2009); therefore, more accurate knowledge about micro-aeration optimization i.e., air intensity (Tsapekos et al., 2017) could help clarify these variations and maximize the potential application of this approach.

In addition to this, the dynamic of the microbial community in micro-aerated AD reactors is also not quite clear. Some studies have reported that micro-aeration has no strong impact on the dynamics of microorganisms (Tang et al., 2004) while other studies have stated that micro-aerated systems have a higher diversity and relative abundance of hydrolytic microorganism, i.e. *Firmicutes*, compared to a strict AD system (Yin et al., 2016). Lim et al. (2014) have reported a 14% enhancement in the relative abundance of *Firmicutes* in a micro-aerated system compared to a strict AD one. In addition to the diversity, methanogenic activity has, on occasion, been found to improve more under micro-aeration conditions in comparison to the conventional AD system (Fu et al., 2016). However, it has also been demonstrated that methanogenic activity inhibition depends more on sulfide concentration than micro-aeration (Jenicek et al., 2011a, 2011b). While Brioukhanov and Netrusov (2007) reported the suppression of SRB growth and activity by oxygen. Jenicek et al. (2011a, 2011b) found that SRB activity was not negatively affected under micro-aeration conditions, and the activity of sulfide-oxidizing bacteria was significantly enhanced. Such a technology is affected by many other factors and these are often referred in the relevant literature (Nguyen and Khanal, 2018). Accordingly, it seems likely that although several studies have been performed to elucidate the role of providing microaerobic environment into AD treating different organic wastes, controversial results have been reported. Furthermore, to our knowledge, the underlying mechanisms of the potential impact of micro-aerated approach in terms of microbial dynamics, methane production, hydrolysis performance and H_2S reduction have not been addressed concurrently. Therefore, a devoted investigation to entirely evaluate and define the potential impact of micro-aeration approach in the entire AD process is necessary.

The current study, therefore, aimed to estimate the role of a micro-aeration strategy in H_2S removal, hydrolysis performance and methane yield enhancement. Chicken manure batch AD reactors were supplemented with different air dosages and the H_2S removal efficiencies and methane yields of the different trials were assessed. 16S rRNA gene sequencing was performed to determine if, and to what extent the micro-aeration strategy would shape the profile of the microbial community structure.

2. Experimental section

2.1. Substrate and inoculum

The substrate used in this study was chicken manure and was obtained from an egg-laying farm belonged to China Agriculture University. The large particles i.e. feathers, were separated and then the manure was sieved to avoid a blockage of tools and kept at 4 °C until usage. The fed chicken manure had a TS, volatile solid (VS), and total chemical oxygen demand (TCOD) of 10%, 7% and 104 g/L respectively. Sludge from a long-term laboratory-scale mesophilic (37 °C) digester fed with chicken manure was used as inoculum, and this had a TS and VS content of 4.3% and 2.2% respectively. The total volatile fatty acids in the inoculum were 300 mg L^{-1} and the total ammonium-nitrogen was 3982 mg L^{-1}. The inoculum digester was operated consecutively for more than 90 days and the inoculum was thus well acclimated through the long term operation. The chemical characteristics of the sludge (inoculum) and chicken manure (substrate) are presented in Table S1 (supplementary material).

2.2. Batch assays experimental setup

All batch experiments were carried out in sealed glass serum bottles with a 70 and 120 mL working and total volume, respectively. The chicken manure and sludge were mixed to reach a substrate/inoculum ratio of 20/80 (volatile solids basis). The batch digesters were instantly flushed with N_2 to provide harsh anaerobic conditions. The butyl rubber stoppers and aluminum seals were utilized to seal the digesters which were at mesophilic conditions (37 ± 1 °C). The micro-aeration treatments were carried out in the first 15 days of digestion using a syringe to inject air. Five sets of micro-aerated reactors were set up: S_1, S_2, S_3, S_4 and S_5 with air intensity of 7, 10, 20, 30 and 50 mL/g VS_{in}, respectively (Table 1). A set of batch reactors were also used without applying micro-aeration (S_0). Additionally, blank digesters (sludge and water) were used to quantify the amount of methane production by inoculum (indigenous respiration). A total of 96 batch digesters were set up (14 replicates for each trial). In order to measure different parameters, two reactors from each trial were terminated every 2–3 days in the first 10 days of the experiment and then every 5 days until the end of the experiment. All the experiments were run until the methane production ceased. The methane production was calculated at standard temperature and pressure conditions.

2.3. Kinetic model

The first-order kinetic model was used as equation (4):

$$\ln (Cs_0 / Cs) = kt \quad (4)$$

where Cs_0 refers to the highest biogas production (mL); Cs stands for the highest biogas production after subtracting the biogas cumulative (mL); k is the rate constant of the first order (day^{-1}); t is the elapsed time (day).

2.4. H_2S and sulfate monitoring

Sulfate in the digestate was measured by Ion Chromatography ICS-3000 (method: HJ/T84-2001). Hydrogen sulfide concentrations were measured (using test kits of Gastec, No, 4H; 104000 ppm) in the reactors with and without air injection to calculate the effect of micro-aeration on H_2S removal (H_2S removal efficiency, R_{H2S}) using equation (5) according to (Choudhury et al., 2019):

$$R_{H_2S} = \frac{H_2S \text{ in control digester (ppm)} - H_2S \text{ in microaerated digester (ppm)}}{H_2S \text{ in control digester (ppm)}} \times 100 \quad (5)$$

2.5. Analytical measurements

2.5.1. Analytical measurements

Total solids (TS), volatile solids (VS), total chemical oxygen demand (TCOD) and ammonia (NH_4^+-N) were analyzed according to standard methods (APHA 2005). Soluble parameters were determined after centrifugation of the samples at 10,000 rpm for 10 min (Cence TGL-16 M, China) and these were then filtered through a 0.22 μm filter. Gas composition was measured with Gas Chromatography (Sp-2100) consisting of a stainless-steel column (10 m × 2 mm) at 0.4 MPa hydrogen pressure and a speed of 60 ml min^{-1}. Inlet, column oven and TCD temperatures were 130, 130, 116 °C, respectively. VFAs were analyzed by Gas Chromatography (Shimadzu GC-2010 plus, Japan) consisting of detector (flame ionization) and column (RTX-wax). The methane values were normalized to standard conditions (1.01 bar and a temperature of 273.2 K). The pH variation was determined using the Mettler-Toledo pH meter.

2.5.2. Microbial analysis

DNA extraction from the samples was performed using the cetyltrimethyl ammonium bromide/sodium dodecyl sulfate method. Universal primers 338F/806R for bacteria sequences and 524F/958R for archaeal sequences were utilized for PCR amplifications. Denaturation at 98 °C (for 1 min, followed by 30 cycles for 10s), annealing at 50 °C (for 30 s), and elongation at 72 °C (for 30 s and then 5 min) were the major properties of the thermal cycling of PCR. The amplicons were sequenced using NEB Next® Ultra™ DNA Library Prep Kit for Illumina (NEB, USA). The Qubit@ 2.0 Fluorometer (Thermo Scientific) with Agilent Bioanalyzer 2100 system and the Illumina HiSeq platform were used for evaluating and sequencing the library quality, and then the reads were merged with FLASH.

The data was explored using the Quantitative Insights Into Microbial Ecology software package (QIIME), and alpha- and beta- were processed using in-house Perl scripts. QIIME quality filters and pick_de_novo_otus.py were also applied for reads filtration and creating an OTU table. Consequently, Chao1, Simpson and Shannon indices were calculated.

3. Results and discussion

3.1. The effect of micro-aeration injection on H_2S mitigation

The cumulative emission of H_2S in biogas when the micro-aeration technology was applied is illustrated in Fig. 1. The cumulative emission of H_2S in the control digesters (10,400 ppm) was

Table 1
Experimental set-up of micro-aeration batch experiment.

	Aeration intensity (mL air/g VS_{in})	Aeration intensity (mL O_2/g VS_{in})	Air dosage (mL reactor $^{-1}$)
S_0	0	0	0
S_1	7.14	1.5	1
S_2	10.7	2.25	1.5
S_3	20	4.2	2.8
S_4	30	6.3	4.2
S_5	50	10.5	7.0

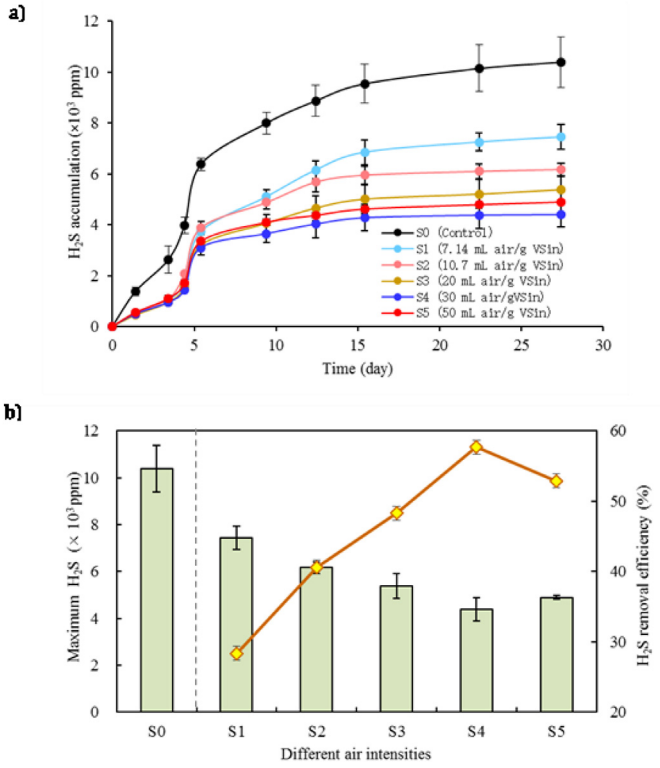


Fig. 1. Accumulative H_2S (a) maximum H_2S and H_2S removal compared to control (b).

used as a baseline to determine the efficiency of H_2S removal in micro-aerated AD. The achieved desulfurization efficiency in micro-aerated AD was between 28% and 58% (Fig. 1b). However, even though the sulfide concentration decreased noticeably, not all the sulfide was removed from the biogas. It is clear, nevertheless, that the accumulation of H_2S decreased with the increase in air intensity. The highest H_2S removal efficiency was observed in micro-aerated AD with an air intensity of 30 mL/g VS_{in}. Conversely, no distinct improvement was observed when air intensity increased to 50 mL/g VSin (S5), indicating that the air intensity at this level may be was over the oxidation capacity of sulfide oxidizers and/or the relative abundance of SRB was not sufficient to utilize all available O_2 and H_2S. Indeed, a slightly reduction of H_2S removal efficiency was even observed at highest air intensity when comparing with air intensity at 30 mL/g VS_{in}, implying that the air was not supplied suitably. This evidence demonstrated that air intensity at 30 mL/g VS_{in} could be used as a crucial threshold to improve H_2S mitigation in the process. Therefore, increasing air intensity up to 30 mL/g VS_{in} helped to enhance the removal efficiency of H_2S by improving the activity of sulfide-oxidizing bacteria. As H_2S is the outcome of SRB activity, this reduction in accumulative H_2S may also be due to SRB inhibition. It has been reported that oxygen at relatively low concentrations could exert toxic effects on sulfate reducers through formation of reactive species and competition with the electron-accepting sulfate, thus blocking dissimilatory sulfate reduction or inactivation of their enzymes (Brioukhanov and Netrusov 2007). The highest H_2S removal efficiency attained in this study (58%) was comparably 2-fold higher than that achieved by Zhou et al. (2007) (Table 2).

On the other hand, the sulfate concentration in the effluent of the micro-aerated digesters was always higher than that found in the control (Fig. 2). Furthermore, the sulfate content rose along with the increase in the air intensity which may be attributed somewhat to sulfide oxidation. The former phenomenon could also be occurred as a result of sufidogenesis limitation, and this meant that sulfate reduction either did not continue occur at all or occurred to a limited extent. Consequently, methanogenic substrates were more available for methanogens. In return, this limitation most probably led to the enhancement of methanogens which had resulted from the H_2S mitigation. In addition, the relative abundance of hydrogenotrophic methanogens, with particular reference to *Methanocellus,* as shown in the next section, increased significantly in the micro-aerated systems. The relationship between SRB and methanogens activities on electron donors has been reported in a previous study (Damianovic and Foresti 2009). Therefore, the increase in the sulfate concentration can be explained by the maintenance of high methanogenic activity, which thus resulted in higher methane yields inside micro-aerated digesters. Remarkably, the sulfate levels in all the micro-aerated reactors (47–326 mg L^{-1}) were far below the inhibition threshold of methanogens. Mizuno et al. (1994) reported methanogenesis inhibition at a sulfate concentration of 1600 mg L^{-1}. However, higher sulfate removal was achieved in the control AD since this removal resulted in the production of H_2S in the biogas, and this could partially explain the higher H_2S content in its biogas (10,400 mg L^{-1}) (Fig. 3). Nevertheless, a dedicated study in continuous reactors to completely monitor the fate of sulfurous compounds and S mass balance in micro-aerated systems is highly necessary.

3.2. Methane yield enhancement with micro-aeration

The methane yields of the control (S_0) and micro-aerated AD (S_1, S_2, S_3, S_4 and S_5) of chicken manure are shown in Fig. 3. As depicted, the methane yields of S_0, S_1, S_2, S_3, S_4 and S_5 were 190, 265, 278, 292, 335 and 199 mL/g VS_{in}, respectively, indicating that the methane yields obviously improved in the micro-aerated AD reactors. A possible reason for this enhancement could be due to the suppression of the inhibition caused by sulfide as discussed above. The highest methane yield was achieved when the air intensity was 30 mL/g VS_{in}, which was 77% higher than that from the digesters in the control AD. The results revealed that the methane yield rose along with the increase in air intensity up to 30 mL/g VS_{in}, implying that the injected air up to that level was maintained below the inhibition threshold. These results are in agreement with Tsapekos et al. (2017) who reported a micro-aeration intensity over 10 mL O_2/gVS may induce the anaerobic process' disturbances. Likewise, Fu et al. (2016) reported that increasing the air load to more than 12.5 mL/L_R-d did not lead to a higher methane yield when corn straw was digested. Subsequently, increasing the air intensity up to 50 mL air/g VS_{added} in this study resulted in a sharp drop in the methane yield. A similar outcome was also observed by Xu et al. (2014), and the reason could be that over-aeration maybe beyond the oxygen consumption and/or anti-oxidation capacity of facultative bacteria which resulted in detrimental effects on methanogens.

In fact, several studies have demonstrated a profound effect on AD process performance when a small amount of air or oxygen is supplemented into the AD process; however, the enhancement achieved in the present study was higher than those achieved before (Table 2). For instance, Tsapekos et al. (2017) reported a 7.2% enhancement in the methane yield of wheat straw when 5 mLO_2/g VS was supplemented in the AD system for 3 days while Lim and Wang (2013) reported a 21% enhancement in the methane yield of brown water and food waste in micro-aerated AD (37.5 mLO_2/L_R-d for 4 days) compared to methane yields attained in a strict AD system (Table 2). These results, and the results attained in this study, highlight how substrate characteristics could have a profound effect on micro-aeration strategy. For instance, chicken

Table 2
Improvement of AD process by micro-aeration technology.

Substrate	Reactor (aeration)	Aeration intensity	Enhancement	References
Wheat straw	Batch (3 day)	5 mL O_2/g VS	7.2% enhancement in methane yield	Tsapekos et al. (2017)
Primary sludge	Batch (per day)	0.00156 vvm[a]	Enhancement on COD degradation	Diak et al. (2013)
Brown water and food waste	Batch (4 day)	37.5 mLO_2/$L_{reactor}$/d	21% enhancement of methane yield	Lim and Wang, (2013)
Corn straw	Batch (per day)	5 mL O_2/g $VS_{substrate}$	16.3% enhancement in methane yield	Fu et al. (2016)
Sludge	CSTR[b] (per day)	13–24 mL O_2/$L_{Reactor}$/d	>99% H_2S removal	Fdz-Polanco et al. (2009)
Synthetic brewery wastewater	UASB[c] (per day)	0.37 L air/$L_{Reactor}$/d	73% H_2S removal	Krayzelova et al. (2014)
Pulp mill wastewater	UASB (per day)	3–6 mL vvm	30% H_2S removal	Zhou et al. (2007)
Chicken manure	Batch (15 day)	7–30 mL air/g VS_{in}	39–77% enhancement in methane yield	This study
Chicken manure	Batch (15 day)	30 mL air/g VS_{in}	58% H_2S removal	This study

[a] vol of air per volume of reactor liquid per minute.
[b] Continuous stirred tank reactor.
[c] Up-flow Anaerobic Sludge Blanket reactor.

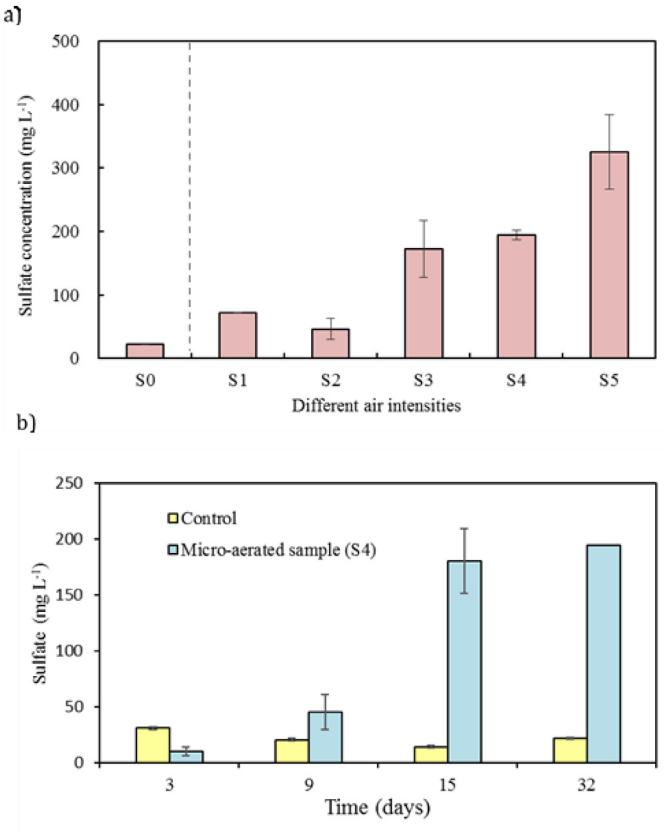


Fig. 2. Sulfate concentration in the digestate (a); time course of sulfate concentration in digestate (b).

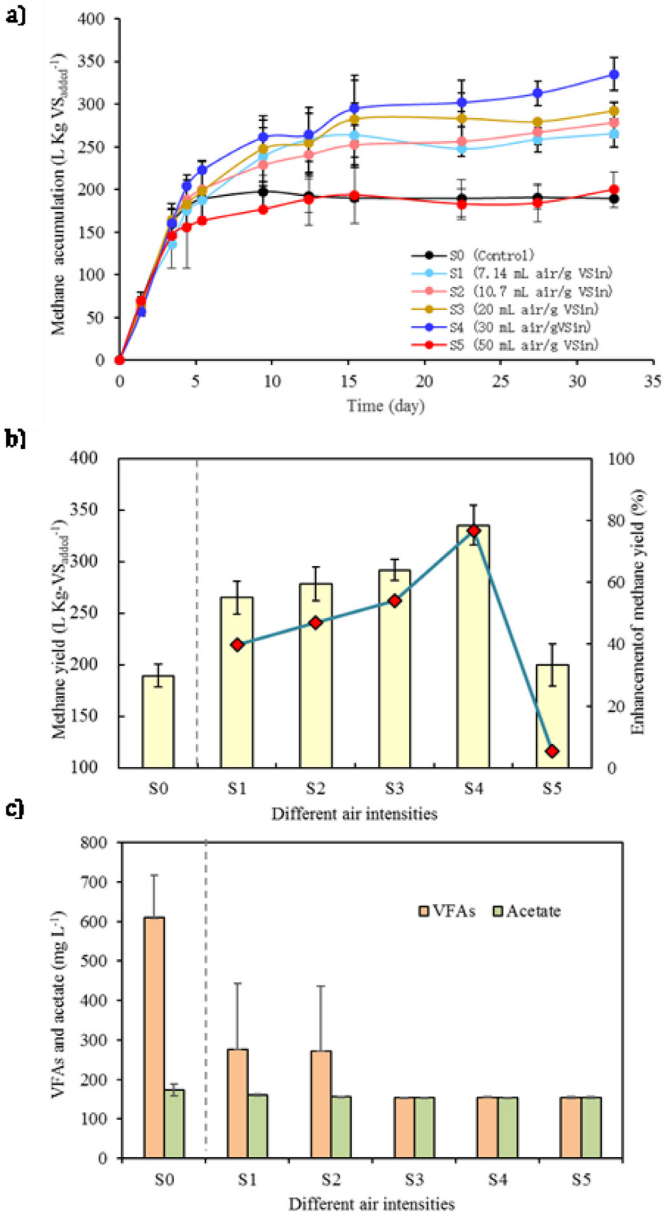


Fig. 3. (a) Time course of methane accumulation (b) Maximum methane yield and micro-aeration effect compared to control AD (c) VFAs and acetate in digestate of control and micro-aerated AD.

manure is rich in both ammonia (as depicted in Table S1, supplementary material) and sulfur-containing amino acids and the synergistic co-inhibition effect of sulfide and ammonia on anaerobic digestion has been recently reported in literature (Surmeli et al., 2017). It is, however, worth noting that the sulfur compounds, NH_3 and VFA levels, the results of degradation of certain substrates, i.e., chicken manure, could provide an appropriate environment for H_2S production and methanogenesis inhibition, and thus the outcome and the positive impact of micro-aeration strategy on the process would vary accordingly which will be more pronounced with such substrates.

Fig. 3c shows the concentration of VFAs and acetate at the end of the digestion time. As depicted, lower VFA concentrations were observed in micro-aerated systems compared to the control AD which was similar to trends reported in previous studies (Jenicek

et al., 2011a, 2011b; Diak et al., 2013). Acetate represented almost all VFAs in the systems with highest air intensities (S_3, S_4 and S_5), indicating that the applied air intensity in these digesters did not provoke a vital shift to undesired higher redox potentials as longer fatty acids, i.e., propionate are preferred in less anaerobic environments with higher redox conditions (Sawatdeenarunat et al., 2015). Conversely to the control digester, as can be seen in Fig. 3, acetate represented all, or at least the highest proportion, of the VFAs in the micro-aerated digesters regardless of air intensity. This also indicated that the micro-aeration procedure promoted the degradation or oxidation of the unusable or less appropriate VFAs from methanogens as concluded by Tsapekos et al. (2017). The current results are in agreement with previous studies that dealt with different substrates in which the application of micro-aeration in the AD process decreased the accumulation of more complex VFAs (Lim and Wang 2013; Tsapekos et al., 2017). It was this that most likely contributed to the higher methane yields attained in the micro-aerated AD compared to the control AD.

In addition, the first-order kinetic model, which is commonly applied to simulate biodegradability as a result of microbial activity in the AD process, was used for comparison between both systems. The value of hydrolysis kinetics obtained by the first-order model demonstrated that the reaction rate was maximized under micro-aerobic conditions (Fig. 4). This verified the enhancement of the methane yield after the micro-aeration strategy as a result of increasing substrate availability to acidogens to produce VFAs for subsequent stages. Acceleration in hydrolysis kinetics resulted in 60% enhancement in hydrolysis efficiency after 4 days in the micro-aerated system (Johansen and Bakke, 2006). This enhancement may be attributed to the enhancement of extracellular enzymes excretion of facultative microorganisms which may have favored the presence of oxygen. More specifically, Gibb's free energy (ΔG^{0-}, KJ/e^-) of different VFAs (including valerate, butyrate, propionate and acetate) conversions are thermodynamically unfavorable (+12.1) and favorable (−106.7) under anaerobic and micro-aerobic conditions, respectively (Nguyen and Khanal, 2018). Therefore, the reason behind this enhancement was that facultative bacteria were most probably favored to shift their metabolic pathway from anaerobic fermentation to aerobic respiration which is more thermodynamically.

3.3. Bacteria diversity, richness, and structure with air injection

The richness and diversity indices of bacteria and archaea communities are shown in Fig. 5. The results indicated that the micro-aeration strategy increased the abundance of the bacteria community as the Choa index was higher in the micro-aerated AD compared to the control AD. Furthermore, the higher Shannon index (4.2 vs 3.8) and the lower Simpson index (0.03 vs 0.05) in the micro-aerated system demonstrated that the microbial community with micro-aerated digesters was also more diversified than that in the control AD system. Indeed, it has been reported that micro-aeration could establish more diverse functional groups (hydrolytic and fermentative bacteria) with higher activity than the strict AD system (Lim et al., 2014; Fu et al., 2016).

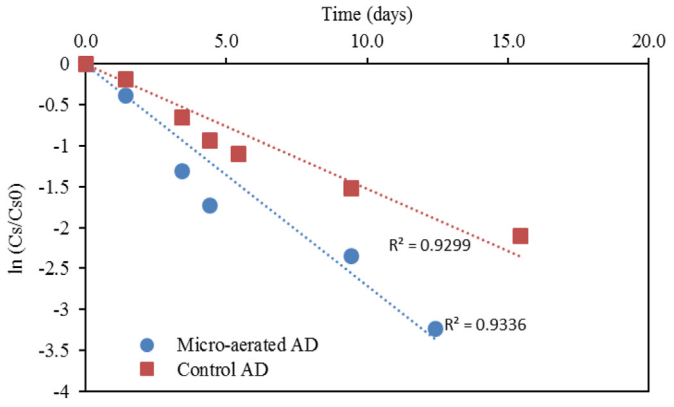


Fig. 4. Methane production kinetics with and without micro-aeration.

In both systems, the most predominant phylum was *Firmicutes*, accounting for 47–48% of total reads, followed by *Bacteroidetes* (36–37%) and *Cloacimonetes* (6–8%) which are the most frequent bacterial sequences reported in the anaerobic digestion process (Lee et al., 2016; Yin et al., 2018). These results differ from that reported by Fu et al. (2016) who demonstrated that the relative abundance of *Firmicutes*, with particular reference to *Clostridiales*, rose under micro-aeration conditions. However, the relative abundance of members belonging to *Proteobacteria* increased by 32% under micro-aerobic conditions. It has been previously reported that the majority of SRB and sulfide-oxidizing bacteria belong to *Proteobacteria* (Krayzelova et al., 2015; Pokorna and Zabranska 2015). Given the fact that the methane yield was enhanced, and hydrolysis kinetics was accelerated in the micro-aerated system, it seems likely that even if there were no variations in the phyla proportions among the systems, bacterial richness and diversity were different, as illustrated above. Furthermore, the bacterial activity may be altered as well. In fact, it has been reported that the growth rate of acetogens decreased considerably under oxidative stress without affecting acetate synthesis activity (Brioukhanov and Netrusov 2007). An anaerobic microenvironment, where the methanogens are protected, could be provided through the oxygen uptake of such bacteria (Kato et al., 1993) as the success of the micro-aeration strategy relies mainly on the oxygen consumption of facultative bacteria that belong to these phyla.

Only those genera with remarkable differences in relative abundance between both systems are shown in Fig. 6a. Among the highly abundant bacteria, *Fastidiosipila*, *Proteiniphilum*, *Alkaliphilus Choacamonas*, *Caldicoprobacter* and *Syntrophomonas* decreased remarkably when shifting from the strict AD to the micro-aerated AD, indicating less tolerance of oxygen. The relative abundance of norank_o_MBAO3, noranck_o_M55-D21, norank_f_*Syntrophomonadaceae* and *Gelria* almost doubled in the micro-aerated system. However, most of them corresponded to uncultured, unclassified, and unranked bacteria and thus their roles in micro-aerated systems were uncertain. *Gelria,* however, was the exception and it has been proposed that species belonging to *Gelria* play a significant role in producing hydrogen, and thereby support their hydrogenotrophic methanogens partners (Lu et al., 2014). In this study, three genera of SRB, namely *Desulfobulbus* and *Desulfovibrio* (*Proteobacteria*) and *Desulfitibacter* (*Firmicutes*), were identified in the control digesters at a relatively low abundance (<1%) and their existence was further decreased by half in the micro-aerated AD. These genera are commonly known as SRB genera in sulfate reduction reactors and belong to the incomplete organic oxidizers (Hao et al., 2014). Species belonging to *Desulfobulbus* and *Desulfovibrio* have been found to utilize ethanol, lactate and hydrogen as electron donors, and sulfate as electron acceptors to produce acetate (incomplete oxidizing SRB) (Liu et al., 2018; Wu et al., 2018). The decrease in relative abundance of H_2-utilized SRB might be compensated by the increase in relative abundance of H_2-utilized methanogens. On the other hand, many species of sulfur oxidizing bacteria, such as *Thioalkalimicrobium*, *Halothiobacillus*, *Paracoccus* and *Pseudomonas* (Friedrich et al., 2001; Sorokin and Kuenen 2005; Vikromvarasiri and Pisutpaisal 2014), were found in the micro-aerated AD at relatively low abundance and these were presumed to be responsible for higher hydrogen sulfide removal efficiency.

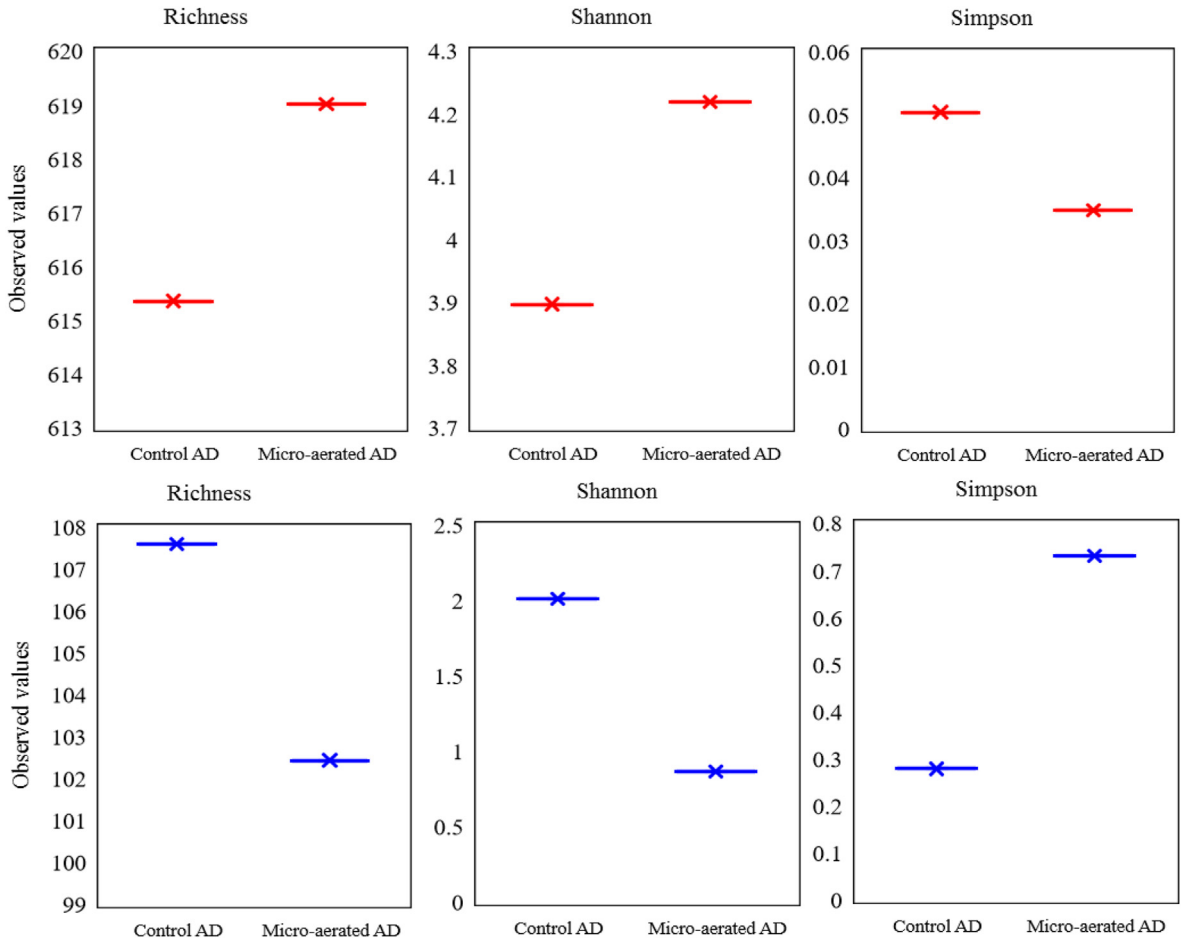


Fig. 5. Alpha diversity indices of the bacterial (red) and archaeal (blue) communities in strict and micro-aerated systems. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.4. Archaea richness, diversity, and structure with air injection

Contrary to the results obtained with the bacteria (Fig. 4), both the richness and diversity of archaea decreased in the micro-aerated AD compared to the control AD (Fig. 4). This indicates that only oxygen tolerant archaea survived in the micro-aerobic regime and thus a more specialized archaea community was assembled. These findings are consistent with a previous study which declared that microbiome clustering was significantly influenced by micro-aeration strategy (Ramos et al., 2014).

The archaeal community is depicted in Fig. 6b, and *Methanoculleus* was the most predominant methanogens in both systems; however, its relative abundance in micro-aerated digesters was 36% higher than that of the control. This result was in accordance with a pervious study (Fu et al., 2016) which found *Methanocelleus* was the prevalent methanogens under microaerobic conditions. *Methanocelleus* is a hydrogenotrophic methanogens which produces methane through the reduction of CO_2 using H_2 as an electron donor (Barret et al., 2013). The increase in relative abundance of such genus most probably led to the rise in hydrogenotrophic methanogenesis rather than hydrogenotrophic sulfidogenesis, resulting in the slight accumulation of sulfate in the micro-aerated system (Fig. 2). This result was in accordance with Tang et al. (2004) who observed that the *Methanoculleus* population increased with micro-aeration strategy when the analysis of 16S rRNA genes and quantitative real-time PCR were applied. The acetoclastic methanogen *Methanoseate*, on the other hand, were found but in relatively low abundance (2.3%) in the control AD, and the decreased (<1%) in the micro-aerated AD. The abundance of hydrogenotrophic methanogens and the lack of acetoclastic methanogens suggests that methane creation was through a two-stage acetate conversion by the syntrophic acetate oxidizing bacteria coupled with hydrogenotrophic methanogens (SAO-HM) (Westerholm et al., 2010). Indeed, even though there is an absence of oxygen tolerance in most methanogens in pure culture, this tendency may be changed in mixed culture such as the culture of anaerobic digestion process. Under mixed culture, facultative bacteria could potentially protect intolerant methanogens from oxygen exposure. Therefore, the enhancement in bacterial diversity and richness that was achieved in the current study (Fig. 5) was the most important contributing factor in methanogens tolerance for oxygen.

4. Conclusion

The simultaneous suppression of sulfide toxicity and the enhancement of the methane yield (77% higher) using micro-aeration technology in order to upgrade the biogas production of chicken manure, were the main findings of the current study. Proper air intensity (30 ml/g VS_{in}) led to the highest H_2S removal (58%) and the maximum methane yield enhancement. The richness and the diversity of bacteria were enhanced and the archaeal *Methanocelleus* exclusively became the dominant methanogens in the new environment. This study supports the proposal that

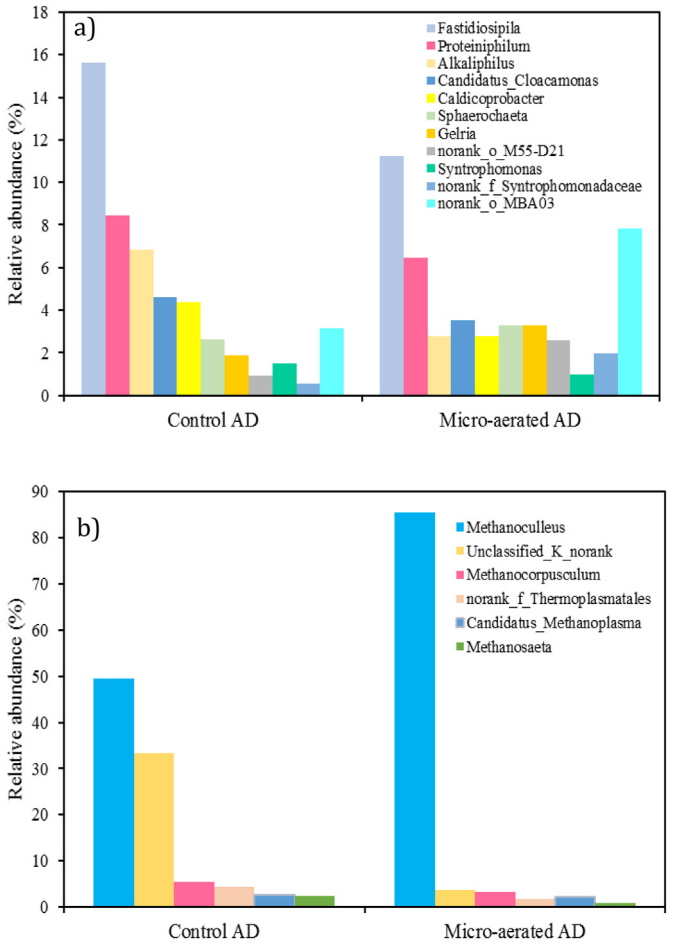


Fig. 6. The relative abundance of the bacterial (a) and archaeal (b) OTUs that had a clear difference between strict AD and micro-aerated AD.

substituting strict AD with micro-aerated AD has the potential to produce cleaner biogas without extra expenditure.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ahmed Mahdy: Writing - original draft, Data curation. **Yunlong Song:** Writing - original draft, Data curation. **Ali Salama:** Writing - review & editing. **Wei Qiao:** Writing - review & editing, Writing - original draft, Formal analysis, Methodology, Conceptualization, Project administration, Funding acquisition, Supervision. **Renjie Dong:** Project administration.

Acknowledgements

This work was partially supported by the National Key Research and Development Program of China (2016YFD0501403) and the Beijing Natural Science Foundation (6182017). Ahmed Mahdy would like to thank State Administration of Foreign Experts Affairs P.R.China. (Project No. WQ20180011) for the financial support.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2020.126687.

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